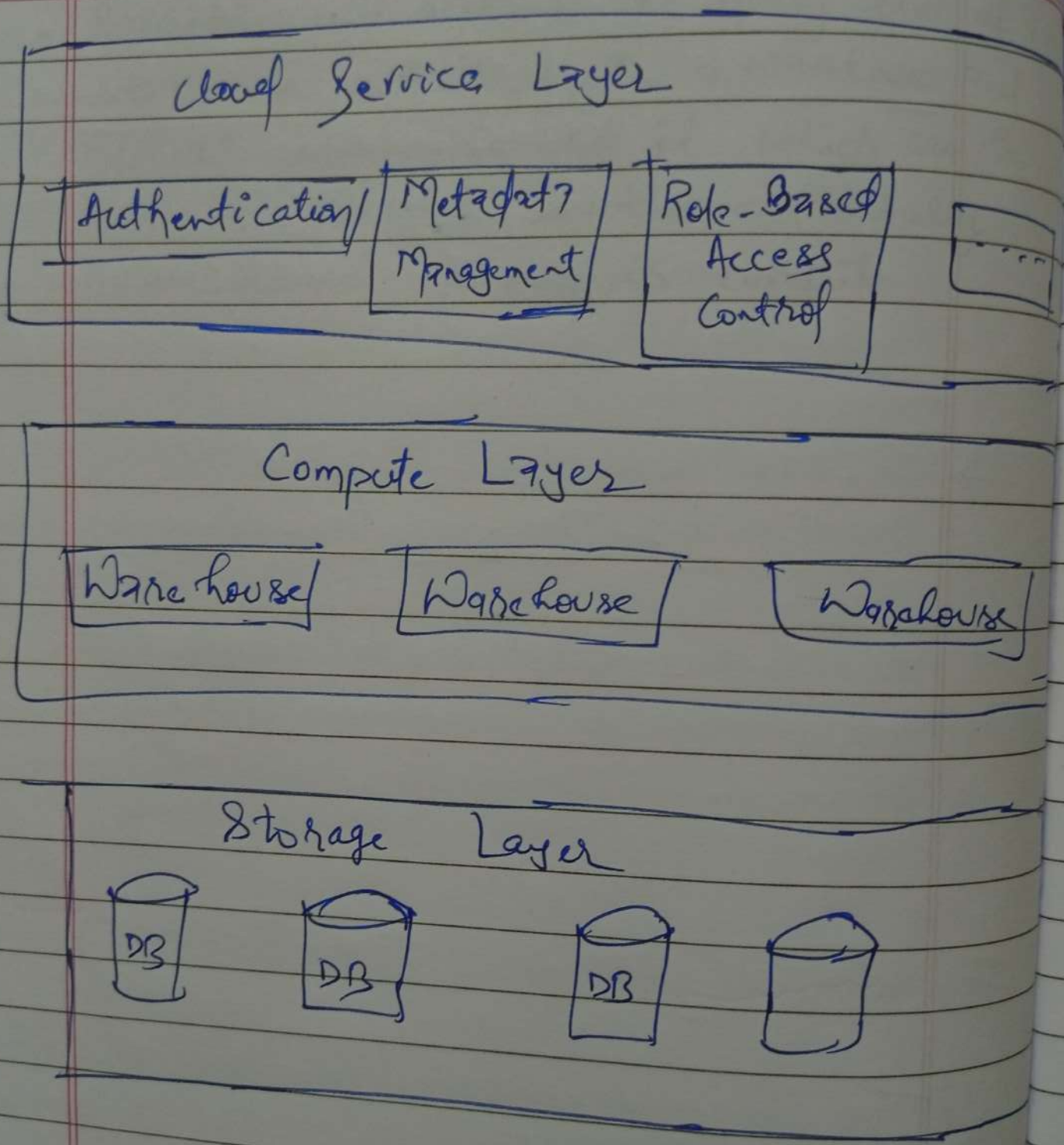
Snowflake Architecture



Snowflake

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Performance Concepts : Virtual Warehouses.

- Virtual Warehouse Introduce
- Warehouse Configuration & Billing : t-sized
- Warehouse State & Properties : started/suspended
- Resource Monitors
- Multi-cluster Warehouses : Scaling out

Performance Concepts : Query Optimization

- SQL Tuning :

ROWS	→ GROUPS	→ RESULT
from	group by	select
join	having	distinct
where		order by
		limit

- Caching :
  - Service layer → Metadata, result cache
  - Warehouse layer → Local disk cache
- Materialized Views
- Automatic Clustering
- Search Optimization Service.

Virtual Warehouse Overview

- A Virtual Warehouse is a named abstraction for a Massively Parallel Processing (MPP) compute layer.

Virtual Warehouses execute :
- DQL operations (SELECT)
- DML operations (UPDATE)
- Data loading operations (COPY INTO)

- As a user we only interact with the named warehouse object not the underlying compute resources.

- Spin up and shut-down a virtually unlimited number of warehouses without resource contention.
- Virtual Warehouse configuration can be changed on-the-fly.
- VW contain local SSD storage used to store raw data retrieved from the storage layer.
- VW are created via the Snowflake UI or through SQL commands.

→ DROP WAREHOUSE my_warehouse;
→ CREATE WAREHOUSE my_med_wh
WAREHOUSE-SIZE = 'MEDIUM';
→ ALTER WAREHOUSE my_wh SUSPEND/RESUME;
suspending a VW puts it in the suspended state, removing the compute nodes from a warehouse.

→ ALTER WAREHOUSE my_wh_2 SET
WAREHOUSE_SIZE = MEDIUM;

→ CREATE WAREHOUSE my_wh_3
MIN_CLUSTER_COUNT = 1
MAX_CLUSTER_COUNT = 3
SCALING_POLICY = STANDARD;

→ CREATE WAREHOUSE my_med_wh
AUTO_SUSPEND = 300;

specifies the no. of seconds of inactivity after which a warehouse is automatically suspended.

By default, auto suspend is enabled and is set to 600 seconds (10 minutes of inactivity)

→ CREATE WAREHOUSE my_med_wh
AUTO_RESUME = TRUE; (default → true)

specifies whether to automatically resume a warehouse when a SQL statement is submitted to it.

→ CREATE WAREHOUSE my_med_wh
INITIALLY_SUSPENDED = TRUE;

specifies whether the warehouse is created initially in the 'suspended' state.

Virtual Warehouse Sizes.

- VW can be created in 10 t-shirt sizes.
- Underlying compute power approximately doubles with each size.
- In general the larger the VW the better the query performance.
- Choosing a size is typically done by experimenting with a representative query of a workload.
- Data loading does not typically require large VW and sizing up does not guarantee increased data loading performance.

Snowflake recommends sticking with X-small, small, M, L unless planning to ingest hundreds or thousands of files concurrently.

VW Billing

X-Small	Small	M	L	XL	2XL	- - -	6XL
1	2	4	8	16	32	64 128	256 512

Credits/hour

0.0003 0.0006

Credits/second

→ On provisioning a VW first 60 seconds are always charged, then on per seconds after that

→ Credit Pricing is determined by region & Snowflake edition.

Resource Monitors

- Resource Monitors are objects allowing users to set credit limits on user managed warehouses.
- Resource Monitors can be set on either the account or individual warehouse level.
- Limits can be set for a specified interval or date range.
- When limits are reached an action can be triggered, such as notify user or suspend warehouse.
- Resource Monitors can only be created by account administrators.

→ CREATE RESOURCE MONITOR analysis_rm
WITH CREDIT_QUOTA = 100
FREQUENTLY = MONTHLY/DAILY/WEEKLY/YEARLY
OR NEVER
START_TIMESTAMP: '2025-10-13 00:00 GMT'
TRIGGERS ON 50 PERCENT DO NOTIFY
ON 75 PERCENT DO NOTIFY
ON 95 " DO SUSPEND
ON 100 " DO SUSPEND_IMMEDIATE;

→ ALTER ACCOUNT SET RESOURCE_MONITOR = analysis_rm;

- Resource Monitor object can be applied at account level and at Warehouse level.

→ ALTER WAREHOUSE 'COMPUTE-WH'
SET RESOURCE-MONITOR = 'analysis-rm';

Multi-cluster Warehouse

Scaling Up: Resizing VW → XSmall → Large

- Scaling up a VW is intended to improve query performance.
- VW can be manually resized via the Snowflake UI or SQL commands.
- Resizing a running warehouse does not impact running queries. The additional compute resources are used for queued and new queries.
- Decreasing size of running warehouse removes compute resources from the warehouse and clears the warehouse cache.

Scaling out: Multi-cluster Warehouses.

- A multi-cluster warehouse is a named group of virtual warehouses which can automatically scale in and scale out based on the number of concurrent users/queries.

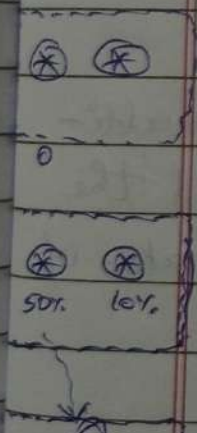
- MIN_CLUSTER_COUNT specifies the minimum number of warehouses for a multi-cluster warehouse.

- MAX_CLUSTER_COUNT specifies the maximum number of warehouses for a multi-cluster warehouse.

Setting these 2 values, the same will put the multi-cluster warehouse in MAXIMIZED mode.

Setting these 2 values differently will put the multi-cluster warehouse in AUTO-SCALE mode.

Standard Scaling Policy



Scaling Out → When a query is queued a new warehouse will be added to the group immediately.

Scaling In → Every minute a background process will check if the load on the least busy warehouse can be redistributed to another warehouse.
If this cond^n is met after 2 consecutive minutes a warehouse will be marked for shutdown.

```
CREATE WAREHOUSE my_mcw_1
  MIN_CLUSTER_COUNT = 1 , MAX_CLUSTER_COUNT = 4
  SCALING_POLICY = STANDARD / ECONOMY;
```

ECONOMY Scaling Policy

Scaling Out → When a query is queued, the system will estimate if there's enough query load to keep a new warehouse busy for 6 minutes.

Scaling In → Every minute a background process will check if the load on the least busy warehouse can be redistributed to another warehouse.
If this cond^n is met after 6 consecutive minutes a warehouse will be marked for shutdown.

Multi-cluster Warehousing Billing

The ~~total~~ total credit cost of a multi-cluster warehouse is the sum of all the individual running warehouses that make up that cluster.

Concurrency Behaviour Properties.

→ CREATE WAREHOUSE my_med_wh
MAX_CONCURRENCY_LEVEL = 6 ; } default - 8 queries

- Specifies the no of concurrent SQL statements that can be executed against a warehouse before either it is queued or additional compute power is provided.

→ CREATE WAREHOUSE my_medium
STATEMENT_QUEUED_TIMEOUT_IN_SECONDS = 60;
(default - no timeout)

specifies the time, in seconds, a SQL statement can be queued on a warehouse before it is aborted.

→ CREATE WAREHOUSE my_medium
STATEMENT_TIMEOUT_IN_SECONDS = 600;

it specifies the time, in seconds, after which any running SQL statement on a warehouse is aborted.

Query Acceleration Service

- The Query Acceleration Service is a feature you can enable on a virtual warehouse.
- Its purpose is to dynamically add serverless compute power to a warehouse when a complex query needs it.

CREATE WAREHOUSE my_wh WITH
ENABLE_QUERY_ACCELERATION = true;

- Snowflake will analyze the query plan of a statement submitted to a Warehouse and offload fragments of it which can be run in parallel to the dynamically requisitioned serverless compute.